INTRODUCTION TO APPLIED NUMERICAL METHODS

The Bisection Method for Roots of Equations

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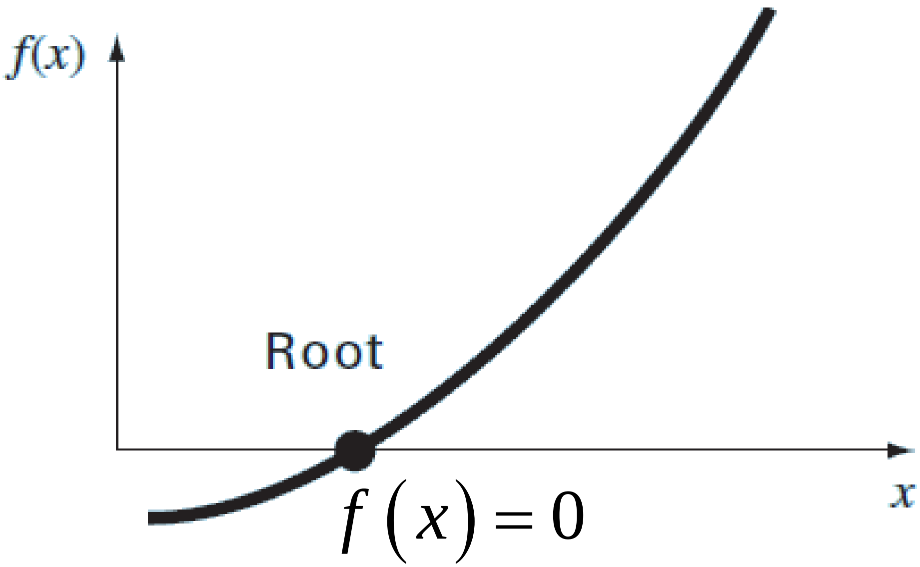
ROOTS OF EQUATIONS

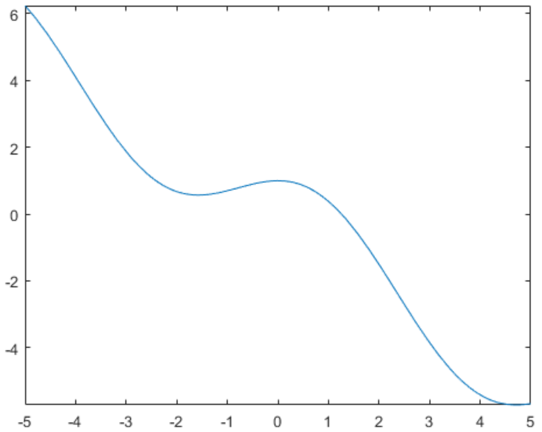


- Quadratic formula

$$ax^2 + bx + c = 0 \quad \Rightarrow \quad x = \frac{-b \mp \sqrt{b^2 - 4ac}}{2a}$$

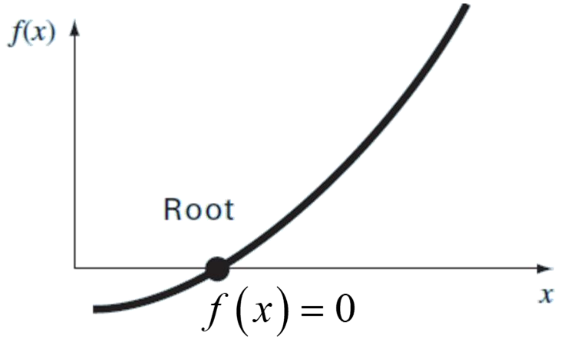
- But

$$ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0 \quad \Rightarrow x = ?$$
$$\sin x + \cos x - x = 0 \quad \Rightarrow x = ?$$
$$e^{-x} - x = 0 \quad \Rightarrow x = ?$$



https://math.stackexchange.com/questions/176583/is-there-a-simple-explanation-why-degree-5-polynomials-and-up-are-unsolvable

OUTLINE OF APPROACHES



Nonlinear Equation Solvers

- **Bracketing**
  - **Bisection**
  - False Position (Regula-Falsi)
- **Graphical**
- **Open Methods**
  - **Newton Raphson**
  - Secant

All Iterative

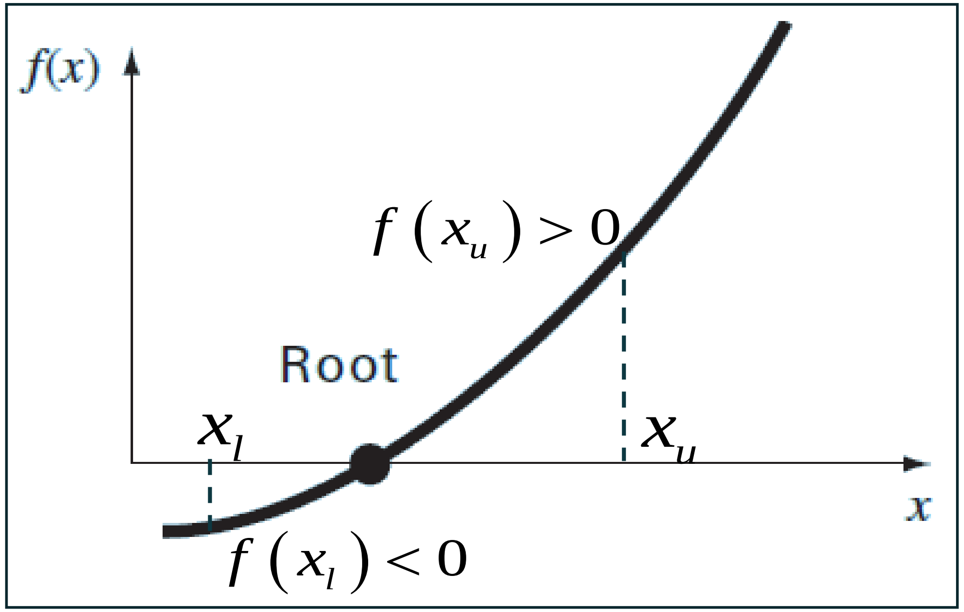
BISECTION METHOD

Principle
Locating an interval where the functional values change signs on both sides

$$f(x_l)\,f(x_u) < 0$$

⬇

At least one real root between x_l and x_u



Numerical Methods for Engineers, 7th Ed.
(S.C.Chapra and R.P.Canale)

Keep halving and searching subintervals successively to locate the sign change and root location!

BISECTION METHOD: THE BISECTION PROCESS

Step 1: Choose lower and upper bounds of x_l and x_u for the root. Ensure the functional value changes sign over the interval, i.e., checking and ensuring

$$f(x_l)\,f(x_u) < 0$$

Step 2: Estimate the root x_r by setting

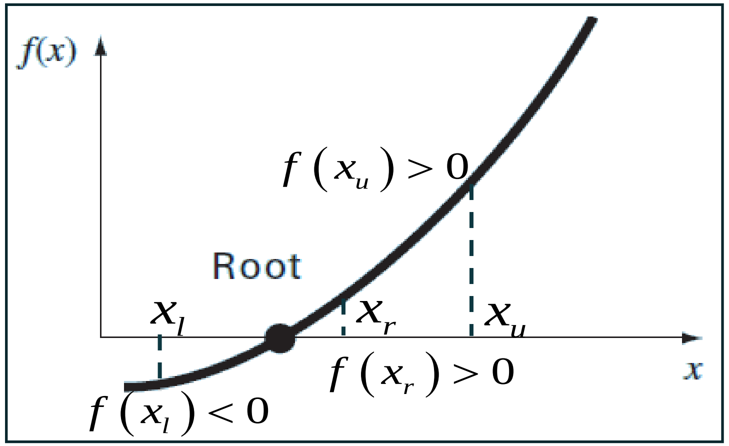
$$x_r = \frac{x_l + x_u}{2}$$

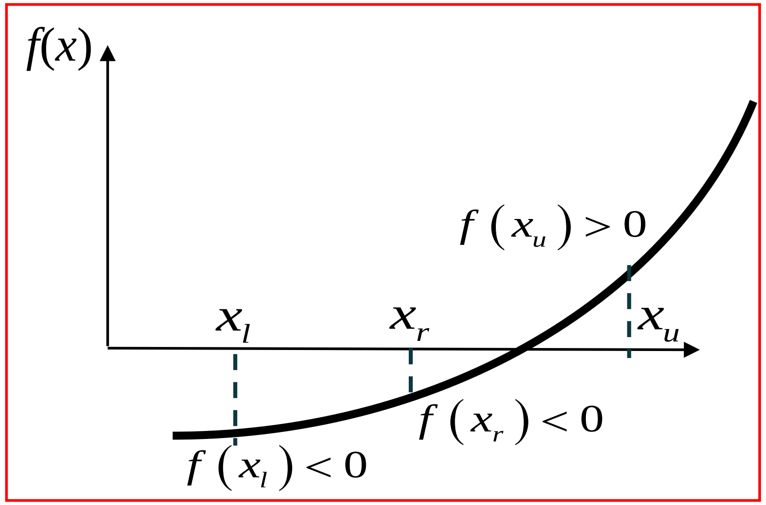
Step 3: Search the subinterval according to sign changes to determine the root location:

(3.1) If $f(x_l)f(x_r) < 0$, the root lies in the lower subinterval, i.e., $x_l<x<x_r$. Therefore, set $x_u = x_r$ and return to step 2.

(3.2) If $f(x_l)f(x_r)>0$, the root lies in the upper subinterval, i.e., $x_r<x<x_u$. Therefore, set $x_l = x_r$ and return to step 2.

(c) If $f(x_l)f(x_r) = 0$, the root equals x_r and terminate the computation.





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BISECTION METHOD

- Given the function $f(x) = x^2 - 1$, the initial bound $x_l = 0$ & $x_u = 3$, please use bisection method to calculate the root.

ITERATION STEPS

- `iter, xl, xu,      xr,    f(xl), f(xu), f(xr), ea`

- 0, 0.000000, 3.000000, 1.500000, -1.000000, 8.000000, 1.250000, 1.000000
- 1, 0.000000, 1.500000, 0.750000, -1.000000, 1.250000, -0.437500, 1.000000
- 2, 0.750000, 1.500000, 1.125000, -0.437500, 1.250000, 0.265625, 0.333333
- 3, 0.750000, 1.125000, 0.937500, -0.437500, 0.265625, -0.121094, 0.200000
- 4, 0.937500, 1.125000, 1.031250, -0.121094, 0.265625, 0.063477, 0.090909
- 5, 0.937500, 1.031250, 0.984375, -0.121094, 0.063477, -0.031006, 0.047619
- 6, 0.984375, 1.031250, 1.007813, -0.031006, 0.063477, 0.015686, 0.023256
- 7, 0.984375, 1.007813, 0.996094, -0.031006, 0.015686, -0.007797, 0.011765
- 8, 0.996094, 1.007813, 1.001953, -0.007797, 0.015686, 0.003910, 0.005848

AN EXAMPLE

The time-dependent velocity, i.e., v(t) of a parachutist who is under the influence of gravity force F_D = mg and the air resistance force F_u = -*cv* is given below

$$\boxed{v = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right)}$$

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Variables	Annotation	Value
m	Parachutist mass	68.1 kg
t	Time	10 s
c	Drag coefficient	??
g	gravity	9.81 m/s^2
v	Velocity	40 m/s

Determine the drag coefficient *c* needed for a parachutist of mass m = 68.1 kg to reach a velocity of 40 m/s @ t = 10 s.

AN EXAMPLE

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Variables	Annotation	Value
m	Parachutist mass	68.1 kg
t	Time	10 s
c	Drag coefficient	??
g	gravity	9.81 m/s^2
v	Velocity	40 m/s

Solution: Define

$$f(c) = \frac{gm}{c}\left(1 - e^{-(c/m)t}\right) - v = 0$$

$$f(c) = \frac{9.81 \times 68.1}{c}\left(1 - e^{-(c/68.1)\times 10}\right) - 40 = \frac{668.06}{c}\left(1 - e^{-0.146843c}\right) - 40$$

True value for comparison: c = 14.8011

AN EXAMPLE

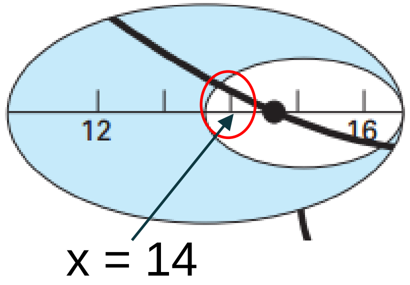
$$f(c) = \frac{668.06}{c}\left(1 - e^{-0.146843c}\right) - 40$$

Step 1: Guess two values of c that give different signs, e.g, x_l = 12 and x_u = 16

$$f(12) = \frac{668.06}{12}\left(1 - e^{-0.146843\times 12}\right) - 40 = 6.114$$

$$f(16) = \frac{668.06}{16}\left(1 - e^{-0.146843\times 16}\right) - 40 = -2.2303$$

Step 2: Estimate the root x_r by setting



$$x_r = \frac{12+16}{2} = 14 \Rightarrow \varepsilon_t = \left|\frac{14.8011 - 14}{14.8011}\right| = 5.4\%$$

Step 3: Search the subinterval according to sign changes

$f(12)\,f(14) = 6.114(1.611) = 9.850 \Rightarrow$ the root lies in the upper subinterval, i.e., 14<x<16

$$x_l = x_r = 14 \qquad x_u = 16$$

Step 4: Estimate the root x_r by setting

$$x_r = \frac{14+16}{2} = 15 \Rightarrow \varepsilon_t = \left|\frac{14.8011 - 15}{14.8011}\right| = 1.3\%$$

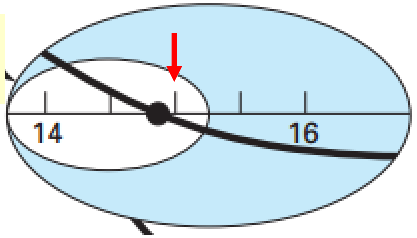
AN EXAMPLE

$$\boxed{f(c) = \frac{668.06}{c}\left(1 - e^{-0.146843c}\right) - 40}$$

Step 5: Search the subinterval according to sign changes

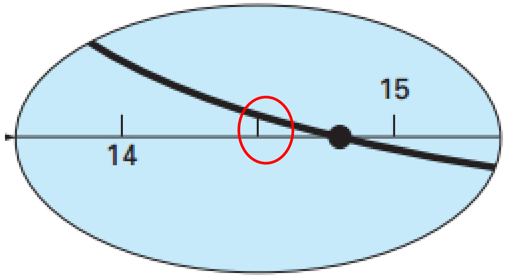
$f(14)\,f(15) = 1.611(-0.384) = -0.619$ ➡ the root lies in the upper subinterval, i.e., 14<x<15

$x_l = 14$ $\quad$ $x_u = x_r = 15$



Step 6: Estimate the root x_r by setting

$$x_r = \frac{14+15}{2} = 14.5$$ ➡ $$\varepsilon_t = \left|\frac{14.8011 - 14.5}{14.8011}\right| = 2\%$$



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TERMINATION CRITERIA AND ERROR ESTIMATE

- When to stop the iteration in the bisection method
  - True error estimate is not available (as the location of the root is not known)
  - Approximate percent relative error ε_a based on the iterative solution will be used

present iteration   previous iteration

$$\varepsilon_a = \left| \frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}} \right| 100\%$$

  - When ε_a becomes less than a prescribed stopping criterion ε_s, the computation is terminated

EXAMPLE CONT'D

(3) Continue the previous example until the approximate error is less than a stopping criterion of ε_s = 0.5%, and compute the relative errors.

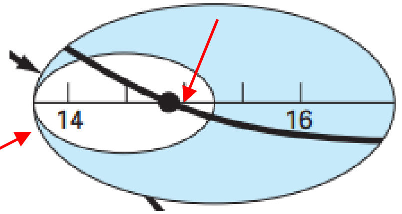
Solution:

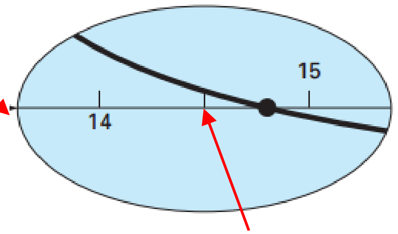
$$\varepsilon_a = \left|\frac{x_r^{new} - x_r^{old}}{x_r^{new}}\right| \times 100\%$$

$$x_{new} = 15, \quad x_{old} = 14 \quad \Rightarrow \quad \varepsilon_a = \left|\frac{15-14}{15}\right| \times 100\% = 6.667\%$$

$$x_{new} = 14.5, \quad x_{old} = 15 \quad \Rightarrow \quad \varepsilon_a = \left|\frac{14.5-15}{14.5}\right| \times 100\% = 3.45\%$$

Iteration	x_l	x_u	x_r	ε_a (%)	ε_t (%)
1	12	16	14		5.413
2	14	16	15	6.667	1.344
3	14	15	14.5	3.448	2.035
4	14.5	15	14.75	1.695	0.345
5	14.75	15	14.875	0.840	0.499
6	14.75	14.875	14.8125	0.422	0.077





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Computing

EVALUATION OF METHOD

<u>Pros</u>

- Easy
- Always find root
- Number of iterations required to attain an absolute error can be computed *a priori*.

<u>Cons</u>

- Slow
- Know *a* and *b* that bound root
- Multiple roots
- No account is taken of $f(x_l)$ and $f(x_u)$, if $f(x_l)$ is closer to zero, it is likely that root is closer to x_l .